

Example 5.1 Write each of the statements using **FOPL**.

(1) All birds are colourful

$$\forall x [\text{Bird}(x) \Rightarrow \text{Colourful}(x)]$$

(2) Every city has a bus which has travelled on all roads in the city.

$$\forall x \{ \text{City}(x) \Rightarrow \exists y \{ \text{Bus}(x, y) \wedge (\forall z) [\text{Road}(x, z)] \Rightarrow \text{Travel}(y, z) \} \}$$

(3) Mary likes all kinds of food.

$$\forall x \text{ Food}(x) \Rightarrow \text{Likes}(\text{Mary}, x)$$

(4) Rice and Bananas are food

$$\text{Food}(\text{Rice}) \wedge \text{Food}(\text{Banana})$$

(5) Anything anyone eats and is alive, is food

$$\forall x \forall y \text{ Eat}(x, y) \wedge \text{Alive}(x) \Rightarrow \text{Food}(y)$$

(6) Varun eats everything Tarun eats

$$\forall x \text{ Eat}(\text{Tarun}, x) \Rightarrow \text{Eat}(\text{Varun}, x)$$

(7) All Romans were either loyal to Caesar or hated him.

$$\forall x \text{ Roman}(x) \Rightarrow \text{Loyal}(\text{Caesar}, x) \vee \text{Hate}(\text{Caesar}, x)$$

(8) People assassinate a ruler they are not loyal to

$$\forall x \exists y \text{ Assassinate}(x, y) \rightarrow \sim \text{Loyal}(x, y)$$

Example 5.2 With correct quantifiers translate to FOPL.

(1) Everyone like apple

$$\forall x, \text{Likes}(x, \text{apple})$$

(2) Someone likes apple

$$\exists x, \text{Likes}(x, \text{apple})$$

(3) All children like apple

$$\forall x, \text{Child}(x) \Rightarrow \text{Likes}(x, \text{apple})$$

(4) Everyone likes apple unless they are allergic to it

$$\forall x, \neg \text{Allergic}(x, \text{apple}) \Rightarrow \text{Likes}(x, \text{apple})$$

(5) Not everyone likes apple

$$\neg(\forall x, \text{Likes}(x, \text{apple}))$$

$$\exists x, \neg \text{Likes}(x, \text{apple})$$

(5) No one likes apple

$$\forall x, \neg \text{Likes}(x, \text{apple})$$

Example 5.3 Translate into FOPL.

(1) Everyone likes some kind of food

$$\forall x \exists y, \text{Food}(y) \wedge \text{Likes}(x, y)$$

(2) There is a kind of food that everyone likes

$$\exists x \forall y, \text{Food}(x) \wedge \text{Likes}(y, x)$$

(3) Someone likes all kinds of food

$$\exists x \forall y, \text{Food}(y) \wedge \text{Likes}(x, y)$$

(4) Every food has someone who likes it

$$\forall x \exists y, \text{Food}(x) \wedge \text{Likes}(y, x)$$

(5) Every successful person has a mentor

$$\forall x \exists y, \text{Person}(x) \wedge \text{Successful}(x) \wedge \text{Mentor}(x, y)$$

(6) Every successful person is a mentor

$$\forall x, \text{Person}(x) \wedge \text{Successful}(x) \Rightarrow \text{Mentor}(x)$$

(7) Every mentor is a successful person

$$\forall x, \text{Mentor}(x) \Rightarrow \text{Person}(x) \wedge \text{Successful}(x)$$

Example 5.13 Translate into FOPL.

(1) Every gardner likes the sun

$$\forall x, \text{Gardner}(x) \Rightarrow \text{Likes}(x, \text{sun})$$

(2) You can fool all the people for some time

$$\forall x \exists y, [\text{Person}(x) \Rightarrow \text{Time}(y) \wedge \text{Fool}(x, y)]$$

(3) You can fool some people all of the time

$$\exists x \forall y, [\text{Person}(x) \wedge \text{Time}(y) \Rightarrow \text{Fool}(x, y)]$$

(4) All pink flowers are smelly

$$\forall x, \text{Flower}(x) \wedge \text{Pink}(x) \Rightarrow \text{Smelly}(x)$$

(5) No pink flower is smelly

$$\forall x, \text{Flower}(x) \wedge \text{Pink}(x) \Rightarrow \neg \text{Smelly}(x)$$

or

$$\neg(\exists x) \text{Flower}(x) \wedge \text{Pink}(x) \wedge \text{Smelly}(x)$$

All students are smart.

$\forall x (\text{Student}(x) \Rightarrow \text{Smart}(x))$

There exists a student.

$\exists x \text{ Student}(x).$

There exists a smart student.

$\exists x (\text{Student}(x) \wedge \text{Smart}(x))$

Every student loves some student.

$\forall x (\text{Student}(x) \Rightarrow \exists y (\text{Student}(y) \wedge \text{Loves}(x,y)))$

Every student loves some other student.

$\forall x (\text{Student}(x) \Rightarrow \exists y (\text{Student}(y) \wedge \neg (x = y) \wedge \text{Loves}(x,y)))$

There is a student who is loved by every other student.

$\exists x (\text{Student}(x) \wedge \forall y (\text{Student}(y) \wedge \neg (x = y) \Rightarrow \text{Loves}(y,x)))$

Bill is a student.

$\text{Student}(\text{Bill})$

Bill takes either Analysis or Geometry (but not both)

$\text{Takes}(\text{Bill}, \text{Analysis}) \Leftrightarrow \neg \text{Takes}(\text{Bill}, \text{Geometry})$

No student loves Bill.

$\neg \exists x (\text{Student}(x) \wedge \text{Loves}(x, \text{Bill}))$

Bill has at least one sister.

$\exists x \text{ SisterOf}(x, \text{Bill})$

Bill has no sister.

$\neg \exists x \text{ SisterOf}(x, \text{Bill})$

Bill has at most one sister.

$\forall x, y (\text{SisterOf}(x, \text{Bill}) \wedge \text{SisterOf}(y, \text{Bill}) \Rightarrow x = y)$

Bill has exactly one sister.

$\exists x (\text{SisterOf}(x, \text{Bill}) \wedge \forall y (\text{SisterOf}(y, \text{Bill}) \Rightarrow x = y))$

Bill has at least two sisters.

$\exists x, y (\text{SisterOf}(x, \text{Bill}) \wedge \text{SisterOf}(y, \text{Bill}) \wedge \neg (x = y))$

Every student takes at least one course.

$\forall x (\text{Student}(x) \Rightarrow \exists y (\text{Course}(y) \wedge \text{Takes}(x,y)))$

Only one student failed History.

$\exists x (\text{Student}(x) \wedge \text{Failed}(x, \text{History}) \wedge \forall y (\text{Student}(y) \wedge \text{Failed}(y, \text{History}) \Rightarrow x = y))$

No student failed Chemistry but at least one student failed History.

$\neg \exists x (\text{Student}(x) \wedge \text{Failed}(x, \text{Chemistry})) \wedge \exists x (\text{Student}(x) \wedge \text{Failed}(x, \text{History}))$

Every student who takes Analysis also takes Geometry.

$\forall x (\text{Student}(x) \wedge \text{Takes}(x, \text{Analysis}) \Rightarrow \text{Takes}(x, \text{Geometry}))$

No student can fool all the other students.

$\neg \exists x (\text{Student}(x) \wedge \forall y (\text{Student}(y) \wedge \neg (x = y) \Rightarrow \text{Fools}(x,y)))$